Three Margins of Electricity Recovery: Coverage, Selective Entry, and Generator Sizing in Japan's Municipal Incinerator Fleet, FY2005–FY2024

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Abstract

Facility counts can misdiagnose electricity recovery when activity, transition, and engineering margins are collapsed. We link 23,593 Japanese municipal- incinerator records for fiscal year (FY) 2005–FY2024 into 1,690 audited stable administrative lineages and 1,767 reported asset episodes. In FY2024, 41.1% of records reported installed capacity, while positive-output facilities handled 80.1% of throughput and installed-capacity facilities held 70.5% of processing design capacity. From FY2005, all-record installed-capacity prevalence rose 19.5 percentage points but only 2.2 points among 732 lineages observed at both endpoints, indicating a strong fleet-composition contribution. First reported installed-capacity entry was sparse: 35 events entered the broad exact-year model. A revision-frozen five-parameter Firth model gave a conditional 300- versus-100 t/day odds ratio of 6.72 (1,999-lineage-bootstrap 95% confidence interval: 4.31–12.46). The 300-t/day level was near the 99th empirical percentile, so support-aware absolute predictions are also reported. Among 6,511 engineering-valid generator-years across 493 lineages, adjusted installed capacity was 79.1%, 58.6%, and 23.5% lower in pre-1990, 1990s, and 2000s cohorts than in 2010-or-later cohorts. A common-control identity decomposition shows that installed sizing is the largest component of every older-cohort log gross-intensity gap. The contribution is a three-margin national diagnosis that separates fleet composition, selective entry, and installed design from annual use; it does not identify physical projects, recoverable potential, or causal effects.

Keywords: municipal solid waste; incineration; waste-to-energy; Japan; generator sizing; capacity factor; administrative record linkage

1 Introduction

Japan's municipal waste system relies extensively on incineration for volume reduction and hygienic treatment. Electricity recovery can use part of the heat released during combustion, but it does not by itself make incineration preferable to prevention, reuse, or recycling. Its value depends on plant design, waste properties, energy use, and the wider waste and electricity systems (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; European Commission, 2017). Japan-specific work likewise extends beyond a binary label of whether a plant generates electricity (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023).

The Ministry's FY2024 national summary reports 415 electricity-generating facilities among 991 incineration facilities, or 41.9% (Ministry of the Environment Japan, 2026). That published context is distinct from this paper's analytical measure: 417 of 1,014 retained FY2024 facility records report positive installed electrical capacity, or 41.1%. The numerator definition and record denominator differ, so the official ratio is not substituted into the analytical calculations.

The most visible national statistic is often a facility count. Yet a small intermittent facility and a large continuous facility each contribute one unit, despite handling different waste volumes. A count describes how widely equipment appears across records, not how much waste passes through generating facilities or where processing capacity is located. Those are distinct coverage questions.

A second problem arises in facility histories. First reporting of installed capacity is uncommon and can occur within an administratively continuous facility, with a reported asset reset, or in a forward-dated record. These cases do not have one verified physical meaning. The workbooks also lack a reliable identifier over the full period: codes are absent in FY2010–FY2012 and fully change between FY2019 and FY2020. Treating them as persistent keys would break histories and create false events.

A third problem concerns gross electricity divided by throughput. This MWh/t ratio combines generator size relative to processing capacity, annual use of installed electrical capacity, and waste loading. Treating it as an independent operating measure can make a design-cohort difference look operational.

This paper addresses the three problems in one national facility-level study. It reconstructs stable administrative lineages before creating lags, separates facility participation from waste-volume and design-capacity coverage, uses a bias-reduced event-history model for sparse first-entry events, and decomposes gross generation intensity into observable engineering components. The aim is descriptive and diagnostic. The study does not estimate the effect of a specific equipment project, infer a physical closure from administrative absence, or calculate net electricity export, useful heat, lifecycle emissions, or recoverable technical potential. Its primary contribution is a three-margin diagnosis: coverage separates equipment counts from covered activity, transition estimates first reported entry among observed non-generators, and components separate installed generator design from annual use.

1.1 Research questions

The three research questions (RQs) are:

> **RQ1: Coverage and fleet composition.** How did the share of facility records reporting installed electrical-generation capacity evolve from FY2005 to FY2024, and how does that count-based measure compare with the share of recorded waste throughput handled by positive-output facilities and the share of waste-processing design capacity located at installed-capacity facilities, and how much endpoint change remains among lineages observed in both endpoint years?
>
> **RQ2: First reported installed-capacity entry.** Among stable administrative lineages first observed without installed electrical-generation capacity, which prior-year age and waste-processing-scale profiles are associated with first reporting positive capacity, and does the evidence change after the risk set is restricted to lineages with positive prior-year waste throughput?
>
> **RQ3: Installed design versus annual use.** Among operating generators, how do raw

installed electrical capacity and annual electrical capacity factor differ across reported start-year cohorts after observed controls; and how do these components combine with waste loading to produce gross MWh/t?

The sequence matters. RQ1 establishes the denominator problem at fleet level. RQ2 then asks which observed non-generating lineages enter the installed-capacity state. RQ3 begins only after positive throughput and gross output are observed, and asks what produces differences within the generating segment. Figure 1 shows the annual count, throughput, and design-capacity coverage measures that anchor this sequence.

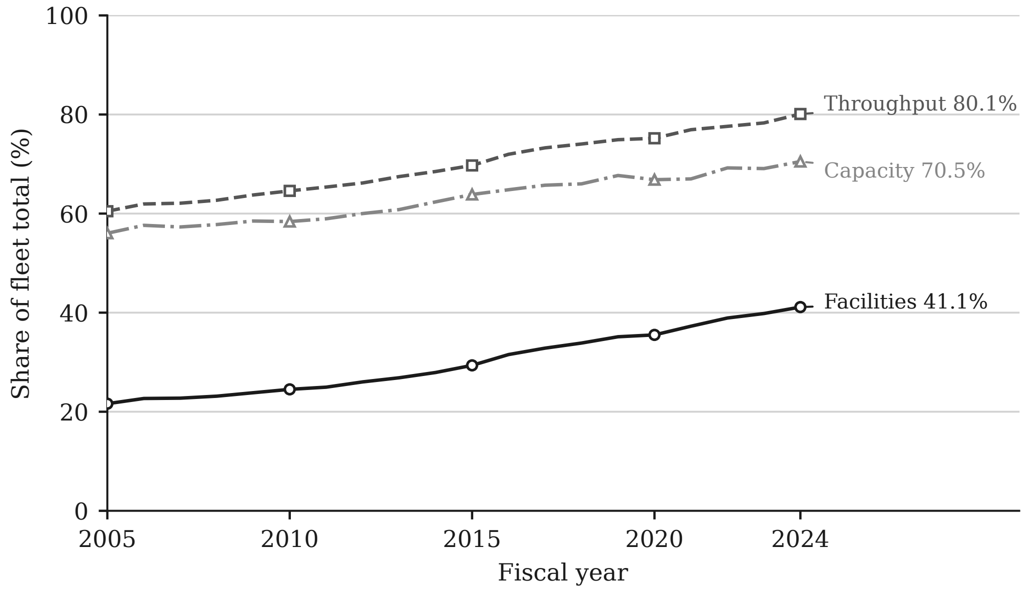

Figure 1: Annual coverage among Japanese incinerator records, FY2005–FY2024. “Facilities” denotes positive-capacity records; “Throughput”, waste at positive-output records; and “Capacity”, processing design capacity at positive-capacity records. Denominators differ.

2 Research Gap and Analytical Contribution

2.1 From one fleet average to three estimands

System studies consider material flows, lifecycle effects, energy substitution, and waste hierarchy (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; Münster & Meibom, 2010). Facility studies benchmark operating plants (Chen et al., 2012; Yeh, 2020), while Japan studies examine policy, power systems, heat use, and technology (Sasao, 2018; Shino, 2019; Tabata & Tsai, 2016; Uno, 2015). These perspectives do not share one denominator or estimand.

The present design separates three estimands. Fleet coverage is an annual ratio whose denominator is either facilities, tonnes, or waste-processing design capacity. Entry is a discrete-time conditional probability among lineages observed at risk. Generator components are conditional associations among positive-throughput, positive-output generator-years that pass stated engineering checks. A coefficient from the generator frame cannot explain why a non-generator enters, and a facility participation rate cannot describe the share of waste processed with electricity output.

This separation permits a low count share and high throughput share to be true simultaneously. It also permits newer cohorts to report higher gross MWh/t because of larger generators even when older generators do not have lower annual capacity factors. Section 3 states that identity explicitly.

2.2 Unresolved questions in the Japanese administrative series

The closest Japanese studies establish that facility scale, operating mode, waste policy, and waste properties matter for heat and electricity output (Sasao, 2018), and that generation per unit waste does not by itself recover the thermal conversion process observed in detailed plant data (Shino, 2019). Those studies do not reconstruct a national facility history across identifier breaks or distinguish later first reporting of installed capacity from outcomes among facilities already generating.

International facility studies evaluate multiple plant activities or revenue performance (Chen et al., 2012; Yeh, 2020). Recent high-profile studies combine richer engineering, environmental, and system data to evaluate hierarchy, effectiveness, or sustainability outcomes (Cui et al., 2026; Han et al., 2025; Liu et al., 2025). The Japanese administrative workbooks used here cannot support comparable optimization, lifecycle, pollutant, or welfare estimands.

They can answer three narrower questions that remain unresolved for this national series: how coverage changes with the denominator, which observed non-generating lineages later first report installed capacity, and whether gross MWh/t differences are expressed through installed sizing or annual capacity use. The contribution is therefore an identity-aware, component-based diagnosis of the observed fleet rather than a new efficiency frontier or investment ranking.

3 Data and Methods

3.1 Source workbooks and recoverable provenance

The source is the Ministry of the Environment Japan General Waste Treatment Survey facility workbook series, also distributed through the Japanese official statistics system (e-Stat, n.d.; Ministry of the Environment Japan, 2026). The study covers 20 fiscal years, FY2005–FY2024. For each year, the checked-in parser reads the first workbook sheet and searches the first six spreadsheet rows for recognized Japanese header terms. The resulting standardized fields include facility and municipality information, reported start year, waste-processing design capacity, annual throughput, furnace and facility attributes, installed electrical capacity, and gross electricity generated.

The provenance audit records all 20 workbooks, their filenames, byte sizes, 256-bit Secure Hash Algorithm (SHA-256) hashes, parsed sheet names, detected header rows, and final field mappings. The files total 15,158,836 bytes and have 20 distinct hashes. Seventeen standardized fields are detected in the older workbooks; 19 are detected from FY2018 onward. Sold-electricity and sales-revenue fields are not detected for FY2005–FY2017, which is one reason the outcome is gross generation rather than net export or revenue. The original download timestamp was not recorded. Volatile checkout modification times are deliberately not persisted; the manifest instead records each workbook's last Git commit timestamp as repository history, not retrieval time. Configured source URLs were not revalidated during the provenance build. The hashes establish which local files produced the results, not the publisher's revision history or a complete acquisition chain.

Parsing yields 23,599 raw rows. Six rows are exact duplicates of another source record and are collapsed before identity resolution, leaving 23,593 unique facility-year records. Derived age is fiscal year minus reported start year. Thirty-six source ages are missing and 355 are negative. Negative values are converted to missing for age-dependent analysis rather than set to zero. Waste-processing utilization is annual throughput divided by 365 times design capacity. Installed generation is indicated by reported electrical capacity greater than zero.

3.2 Stable administrative lineages and asset episodes

Official facility codes cannot support the longitudinal design by themselves. All 3,716 rows in FY2010–FY2012 lack a code. Between FY2019 and FY2020, both years contain codes, yet the sets have zero overlap. The latter break is a complete recode, not evidence that the entire fleet was replaced. Across that transition, the identity resolver restores 1,064 adjacent-year lineage links, equivalent to 97.3% of FY2019 records. Across the longer FY2009–FY2013 bridge, 882 official codes overlap while 1,135 stable lineages are linked.

The resolver therefore constructs a *stable administrative lineage*: a sequence of records judged to describe the same continuing administrative facility or site history. The implementation field is `stable_site_id`, but the term does not assert that ownership, buildings, furnaces, or other physical assets remain unchanged. The algorithm is deterministic and proceeds as follows.

1. Text and identifiers are normalized conservatively, including Unicode form, spacing, punctuation, and numeric-code formatting. A complete standardized row fingerprint identifies exact duplicate records, which are collapsed before matching.
2. Records are compared only within prefecture. Candidate evidence combines normalized facility name, municipality, reported start year, waste-processing capacity, furnace count, facility type, and the annual official code. Code agreement is supporting evidence, but contradictory name and configuration evidence can veto a code match.
3. Adjacent fiscal years are resolved before gaps, preventing an older history from winning a tie over an otherwise equivalent immediately prior record. Short gaps of up to four years are considered afterward. Within each prefecture-year problem, one-to-one global assignment prevents one prior record from being assigned to multiple current records.
4. Unmatched records seed new deterministic lineage IDs from canonical record fingerprints rather than row order. Within a lineage, a new asset episode is created when reported start year changes materially in either direction, a mature record resets to a near-zero age, or a major name and configuration discontinuity appears.

The result contains 1,690 stable administrative lineages and 1,767 asset episodes, with no duplicate lineage-year and no history longer than the 20-year study window. The match audit classifies 15,324 records as code plus exact-name links, 274 as code with other supporting evidence, 6,204 as exact-name links without code support, 101 as fuzzy-name links without code support, and 1,690 as new lineage seeds.

Candidate links below the minimum score are excluded before assignment, as are ambiguous links lacking an exact-name or official-code signal. This removes 3,092 sub-threshold and 15,308 weak ambiguous candidate edges. Sixteen accepted links, spanning 14 lineages, still have a low current-record or prior-record margin but retain strong evidence; every one is exposed in the identity audit.

The entry and component analyses therefore include whole-lineage identity-certain sensitivities that exclude all 14 affected lineages rather than selectively removing individual years.

Known difficult cases are embedded as executable checks. Three pairs that must link and three pairs that must remain separate are tested directly, including examples from the FY2019–FY2020 recode and earlier code reuse. The resolver is also rerun after random row permutation and after insertion of an unrelated synthetic record in six difficult or duplicate-bearing prefectures. The lineage/episode mapping must remain unchanged. These golden-link, separation, permutation, and insertion tests reduce identifiable implementation risks; they do not make probabilistic record linkage certain. Any result depending on a small number of lineages remains vulnerable to unresolved name or configuration ambiguity. The uncertainty flag makes that vulnerability testable rather than implicit.

A blinded clerical-review packet provides a further validation gate consistent with guidance for linked administrative cohorts (Harron et al., 2020). It contains all 35 modeled event links, all 16 accepted uncertain links, all 31 gap links, 50 deterministic FY2019–FY2020 bridge links, and a stratified sample of other accepted pairs. Scores, algorithmic decisions, and final lineage IDs are withheld. Until a second reviewer completes the packet and disagreements are adjudicated, it is a protocol and sensitivity resource rather than evidence of independent validation. Completion remains a journal-submission gate.

3.3 Analytical frames and estimands

The three RQs use related but non-identical samples. Table 1 prevents their denominators from being blended.

Table 1: Analytical frames after identity and data-quality checks

Frame	Facility-year rows	Stable lineages	Events	Estimand
Full administrative panel	23,593	1,690	N/A	Annual facility, throughput, and design-capacity coverage
Descriptive non-generator risk set	16,519	1,223	55	Observed first installed-capacity entries during follow-up
Broad exact-year complete-covariate entry model	15,154	1,137	35	Conditional annual administrative-lineage entry odds using prior-year covariates
Exact-year model after prior operation	13,072	1,019	33	Nested sensitivity requiring positive prior-year throughput
Same-asset-episode continuity sensitivity	15,095	1,135	24	Entry odds after excluding transitions that cross an inferred asset-episode boundary
Identity-certain-lineage sensitivity	15,107	1,130	35	Broad model after excluding every lineage containing an accepted uncertain identity link
Positive-throughput, positive-output generators	6,660	504	N/A	Descriptive operating-generator frame
Engineering-valid component model	6,511	493	N/A	Conditional generator-component associations

The entry sample excludes 467 lineages already reporting positive installed capacity in their first observed year because their entry time is left-censored. A remaining lineage contributes annual risk rows until its first positive capacity observation or its last observed non-generating year. The first observed risk row is dropped from modeling because a lagged predictor is not available. The complete-covariate frame further drops 120 rows with missing lagged age or processing capacity, and 22 non-adjacent rows are excluded from the exact-year model; none of those 22 rows contains an event.

Separately, an audit classifies all 55 descriptive events from administrative continuity, reported resets, and forward-dated records. These pathway labels do not verify physical project mechanisms. Their exploratory outcome comparison is reported only in the supplement.

3.4 RQ1: coverage definitions and fleet composition

For fiscal year t, let N_t be all retained facility records, I_{it}^K indicate positive installed electrical capacity, I_{it}^G indicate positive gross generation with positive throughput, W_{it} be annual throughput, and C_{it} be waste-processing design capacity. The three headline coverage measures are

$$P_t^{\text{facility}} = \frac{\sum_i I_{it}^K}{N_t}, \tag{1}$$

$$P_t^{\text{throughput}} = \frac{\sum_i W_{it} I_{it}^G}{\sum_i W_{it}}, \tag{2}$$

and

$$P_t^{\text{design}} = \frac{\sum_i C_{it} I_{it}^K}{\sum_i C_{it}}. \tag{3}$$

The first uses facility records, the second tonnes, and the third tonnes per day of design capacity. Positive output is used in the throughput numerator because installed capacity does not guarantee positive annual generation. For the engineering-valid subset, fleet gross intensity has the exact decomposition

$$\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}} = \left(\frac{\sum_i W_{it}^{\text{valid}}}{\sum_i W_{it}}\right)\left(\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}^{\text{valid}}}\right). \tag{4}$$

The first term is valid generator-throughput coverage and the second is throughput-weighted conditional gross intensity. This identity distinguishes how much waste is covered from how much gross electricity is reported per tonne within the covered segment.

Annual prevalence is a repeated cross-section. A composition diagnostic compares all endpoint records with stable lineages observed in FY2005 and FY2024, a same-reported-episode subset, and lineages observed in all 20 years. Endpoint-only lineages are also compared, but administrative appearance and disappearance are not interpreted as physical openings or closures. The groups have different denominators and are not an additive causal decomposition.

3.5 RQ2: sparse first-entry model

The event is the first observed year with positive installed electrical capacity after a lineage has been observed without it. It is an administrative state transition. It does not uniquely identify first physical operation, equipment installation, or a particular construction history.

Discrete-time event-history analysis represents each at-risk lineage-year as a binary outcome (Allison, 1982; Beck et al., 1998). For lineage i in year t, the revision-frozen primary model is

$$\operatorname{logit}\{\Pr(Y_{it}=1 \mid Y_{i,t-1}=0)\} = \alpha+\beta_A\frac{A_{i,t-1}}{10}+\beta_C\log\left(1+\frac{C_{i,t-1}}{100}\right)+\beta_T\frac{t-2014.5}{5}+\beta_R\log(1+R_{it}). \tag{5}$$

Here, A is prior-year reported facility age, C is prior-year processing design capacity in t/day, and R is elapsed observed time at risk. Age is scaled per ten years and calendar time per five years. Including the intercept, the specification has five parameters for 35 broad-frame events. A dated decision memo records the model before the revised fit; it was not externally preregistered. The earlier 11-parameter age-band, calendar-era, and duration- band model is retained as sensitivity.

Ordinary maximum-likelihood logit is vulnerable to small-sample bias and separation in a sparse event design. The primary estimator therefore uses Firth's bias reduction (Firth, 1993) and its finite-estimate solution to separation (Heinze & Schemper, 2002), maximizing the penalized log-likelihood

$$\ell_F(\theta) = \ell(\theta) + \frac{1}{2}\log\left|\mathcal{I}(\theta)\right|, \tag{6}$$

where ℓ is the binomial log-likelihood and $\mathcal{I}$ is the expected information matrix. The Jeffreys-prior penalty reduces first-order bias and keeps finite estimates under separation. The coefficient table labels the fitted-model standard errors, confidence intervals, and term-level p-values as model-based. To represent repeated observations from the same lineage, the primary uncertainty calculation uses 1,999 deterministic cluster-bootstrap replications that resample complete lineages with replacement. Percentile intervals use the resulting coefficient distributions. Every requested replication must converge and return all focal coefficients; otherwise the analysis fails rather than silently dropping a bootstrap draw.

Four frames are fitted. The broad exact-year frame includes every complete at-risk row whose lag belongs to the same stable administrative lineage; it can cross an inferred asset-episode boundary because its estimand is first administrative-lineage entry. The prior-operation frame is a nested sensitivity that additionally requires positive prior-year throughput. It is not an independent comparison group, so differences between it and the broad frame are not treated as an equality or equivalence test. A same-asset-episode continuity sensitivity excludes the 59 exact-year rows that cross an inferred episode boundary, including 11 events. An identity-certain sensitivity excludes every lineage containing any of the 16 accepted uncertain links. These two additional frames show whether inference depends on the continuity and linkage rules.

For an intuitive scale contrast, the odds ratio comparing 300 with 100 t/day is

$$
\begin{aligned}
OR_{300:100} &= \exp[\beta_C \{\log(1 + 300/100) - \log(1 + 100/100)\}] \\
&= \exp(\beta_C \log 2).
\end{aligned} \tag{7}
$$

Because 300 t/day lies near the 99th empirical percentile, standardized annual probabilities are also reported at 24, 60, and 120 t/day, the risk-frame 25th percentile, median, and 75th percentile. Predictor correlations and variance inflation factors delimit interpretation of the calendar and elapsed-risk terms.

The earlier 11-parameter Firth model and conventional logit and complementary-log-log fits are retained as specification sensitivities.

3.6 RQ3: engineering decomposition and component models

Let G_{it} be annual gross generation in MWh, W_{it} annual throughput in tonnes, K_{it} installed electrical capacity in kW, and C_{it} waste-processing design capacity in t/day. Define

$$
Y_{it} = \frac{G_{it}}{W_{it}} \qquad \text{(gross generation intensity, MWh/t),} \tag{8}
$$

$$
D_{it} = \frac{K_{it}}{C_{it}} \qquad \text{(generator design intensity, kW per t/day),} \tag{9}
$$

$$
F_{it} = \frac{G_{it}}{8.76 K_{it}} \qquad \text{(annual electrical capacity factor),} \tag{10}
$$

$$
U_{it} = \frac{W_{it}}{365 C_{it}} \qquad \text{(waste-processing utilization).} \tag{11}
$$

The factor 8.76 converts one kW operating for 8,760 hours into annual MWh. These definitions imply the exact facility-year identity

$$
Y_{it} = \frac{8.76}{365} \frac{D_{it} F_{it}}{U_{it}} = 0.024 \frac{D_{it} F_{it}}{U_{it}}. \tag{12}
$$

Gross MWh/t is thus a ratio produced jointly by installed sizing, annual use of that electrical capacity, and annual waste loading. The identity does not say that any one component is exogenous. For example, throughput, capacity factor, maintenance, and waste composition may be jointly determined during a year. The derived quantities also share numerators and denominators. Associations or changes in model fit among them are therefore partly mechanical and are used to describe reported structure, not to identify an engineering mechanism.

The primary sample requires positive installed capacity, throughput, and gross output. Values are excluded rather than clipped if gross intensity falls outside 0.01–0.80 MWh/t, electrical capacity factor outside 0.02–1.20, waste-processing utilization outside 0.02–1.20, or generator design intensity outside 0.1–100 kW per t/day. A non-missing non-negative reported age and all model fields are also required. Of 6,660 positive-throughput, positive-output rows, 149 fail at least one check, leaving 6,511 rows across 493 lineages. Heating value between 3 and 25 megajoules per kilogram (MJ/kg) is audited as a plausibility field but is not an inclusion condition for the primary component models; 102 otherwise valid rows lack it.

Two pooled component models are estimated by ordinary least squares with standard errors clustered by stable lineage (Wooldridge, 2010). Installed electrical capacity is modeled in its raw reported unit:

$$\log K_{it} = \alpha_K + V_{it}'\beta_K + \beta_{KC}\log C_{it} + T_{it}'\eta_K + \lambda_t + \varepsilon_{Kit}, \tag{13}$$

$$\begin{aligned} \log F_{it} = \alpha_F + V_{it}'\beta_F + \beta_{FC}\log C_{it} + \beta_U U_{it} \\ + T_{it}'\eta_F + \lambda_t + \varepsilon_{Fit}. \end{aligned} \tag{14}$$

V_{it} contains reported start-year cohorts before 1990, 1990–1999, and 2000–2009 relative to 2010 or later. Reported start year is an administrative design-vintage marker, not a verified boiler or generator installation date. T includes furnace count and coarse furnace and facility-type groups; λ_t are fiscal-year indicators. Because $\log D = \log K - \log C$, the corresponding design-intensity elasticity is $\beta_{KC} - 1$ under identical controls; this is an algebraic translation, not independent corroboration. A direct gross-output model replaces $\log C$ with $\log W$ and $\log K$.

For relative component attribution, additional regressions use one identical control matrix for $\log D$, $\log F$, $\log U$, and $\log Y$. The accounting identity and ordinary least-squares linearity imply for every cohort contrast

$$\gamma_Y = \gamma_D + \gamma_F - \gamma_U. \tag{15}$$

This common-control decomposition is separate from the primary capacity-factor model, which conditions on U and answers an equal-utilization comparison.

A specification diagnostic compares a legacy-style regression of $\log Y$ on reported age, processing scale, waste utilization, heating value, technology, and year with the same model after adding $\log D$. Both specifications use the 5,806 engineering-valid rows with reported heating value in the 3–25 MJ/kg plausibility range. This is not a causal mediation design. It asks whether coefficients previously attributed to age, scale, or utilization are stable after the omitted installed-sizing component is represented.

Robustness checks split the period into FY2005–FY2014 and FY2015–FY2024, give each lineage equal total weight, and apply conservative and broad predefined engineering bounds. Asset-episode fixed effects and exact-adjacent first differences are used only for operating components that vary meaningfully within an episode. Design intensity is predominantly a between-asset attribute, so within-episode change is not presented as its primary estimand.

4 Results

4.1 RQ1: facility counts understate waste-volume coverage

The official 415/991 ratio (41.9%) describes the Ministry's published count of electricity-generating facilities. The results below use the separate analytical definition of positive installed capacity among retained facility records.

Annual repeated-cross-section installed-capacity prevalence rises from 21.6% of facility records in FY2005 to 41.1% in FY2024. Positive-output facilities handle a much larger share of waste: throughput coverage rises from 60.5% to 80.1% over the same period. The design-capacity share at installed-capacity facilities moves from 56.0% to 70.5%. Figure 1 therefore shows long-run increases with year-to-year variation under all three denominators, alongside a persistent count-volume gap.

That 19.50-percentage-point increase is largely compositional. Among 732 lineages observed at both endpoints, prevalence rises only from 29.92% to 32.10% (2.19 points); among 678 endpoint-common lineages retaining the same reported episode, it rises 0.88 points. FY2024-only lineages report 64.54% participation, versus 11.26% among FY2005-only lineages. The trend therefore does not describe widespread conversion among continuing facilities, and the endpoint-only groups are not verified openings or closures.

The FY2024 cross-section makes the distinction concrete. The administrative panel contains 1,014 facility records, of which 417 report positive installed electrical capacity, giving the 41.1% participation rate. There are 410 positive-throughput, positive-output facilities. They process 24.70 million of the 30.84 million recorded tonnes, or 80.1%. In contrast, 469 operating non-generators process 6.14 million tonnes, or 19.9%. Another 126 records report neither positive throughput nor generation, and nine installed-capacity records report no positive output. Because output and installed-capacity flags are not identical, these segment counts should not be forced into a single binary partition.

After the predefined output and capacity-factor checks, valid generator rows cover 79.7% of FY2024 throughput. Their throughput-weighted conditional gross intensity is 0.425 MWh/t. Multiplying 0.797 by 0.425 gives 0.338 MWh per total fleet tonne, exactly matching valid gross generation divided by all recorded throughput. This arithmetic is more informative than the facility count alone: most recorded waste is already processed at facilities reporting positive generation, even though installed equipment appears in fewer than half of facility records.

The result does not imply that the remaining 19.9% of operating throughput can or should all be redirected or equipped. It also does not measure electricity used internally, net export, heat recovery, marginal emissions, or project cost. Its contribution is denominator discipline: 41.1%, 80.1%, and 70.5% answer different questions, and no one of them is a sufficient measure of the remaining opportunity.

4.2 RQ2: entry is rare and strongly associated with processing scale

The descriptive risk set contains 55 first reported installed-capacity entries. The 35 exact-year modeled events comprise 24 continuity-lineage and 11 rebuild/replacement-like entries; none is forward-dated. The four calendar eras contain 7, 4, 14, and 10 modeled events, respectively. In the broader bridge, 47 of 55 descriptive events show positive gross generation in the event year and 51 do so by the following observed fiscal year.

Only 35 events remain after requiring an exact one-year lag and complete prior age and capacity; 33 remain after requiring positive prior-year throughput. This sparse count is central to interpretation. The Firth model addresses finite-estimate and first-order bias problems, but it cannot create information that the panel does not contain.

Observed rates already show strong scale ordering. Entry occurs in 1 of 3,854 risk rows in the smallest processing-capacity quartile (0.026%), 2 of 4,175 in the second quartile (0.048%), 9 of 3,702 in the third (0.243%), and 23 of 3,423 in the largest (0.672%). Because risk duration, calendar period, and age differ across these groups, the adjusted model is needed before interpreting the gradient.

In the revision-frozen five-parameter broad Firth model, the coefficient on $\log(1 + C/100)$ is 2.749. The implied 300-versus-100 t/day odds ratio is 6.72 (1,999-lineage-bootstrap 95% CI 4.31–12.46). The prior-operation, same-episode, and identity-certain odds ratios are 7.09 (4.08–13.76), 7.15 (4.44–14.05), and 6.76 (4.23–12.30), respectively.

Standardized annual predictions are 2.53 and 16.66 entries per 1,000 facility- years at 100 and 300 t/day. The latter is a tail prediction: 300 t/day is at the 98.98th empirical percentile, with 315 risk rows and 4 events at or above it. Predictions at 24, 60, and 120 t/day are 0.68, 1.37, and 3.29 per 1,000. Calendar time and elapsed risk are correlated ($r = 0.909$; variance inflation factors 5.76 and 6.15), but the processing-scale variance inflation factor is 1.10. The temporal coefficients are therefore not interpreted separately.

The broad continuous-age coefficient is -0.327 per ten years (bootstrap CI -0.774 to 0.070); the prior-operation and identity-certain intervals also span zero. The same-episode estimate is more negative (-0.751; CI -1.364 to -0.206) but relies on only 24 events. Age is therefore continuity-sensitive rather than a general equipment-age response. The earlier 11-parameter model supports the same caution.

Reclassifying each event leaves scale odds ratios from 6.12 to 7.30; deleting each event's entire lineage leaves 6.13 to 7.30. No one modeled event or event lineage creates the scale result. These perturbations diagnose influence; they do not validate physical mechanisms.

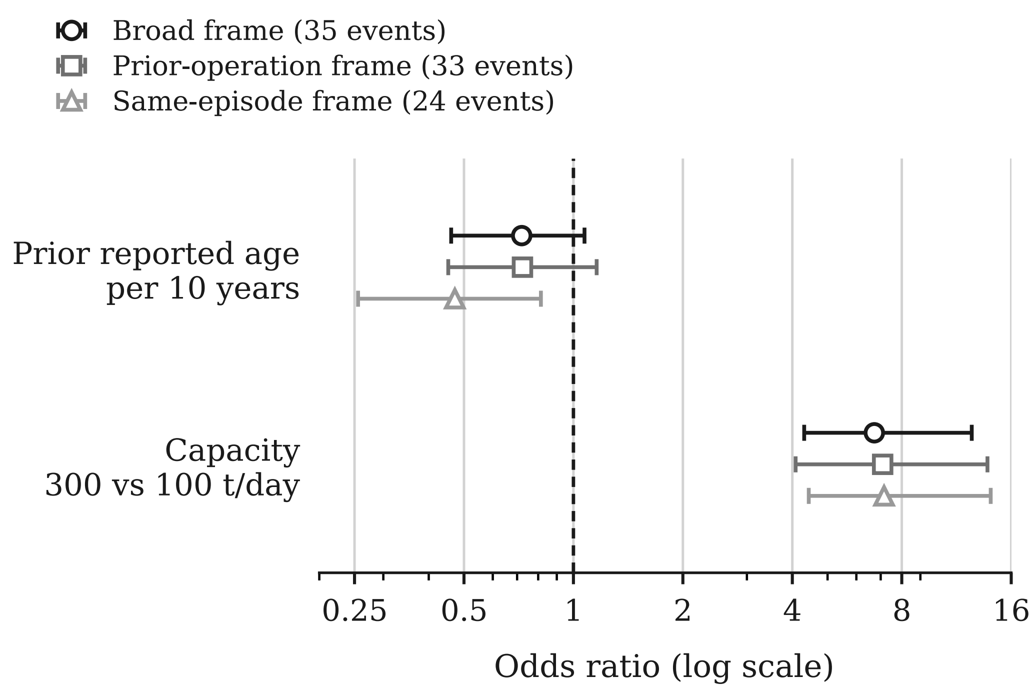

Figure 2: Five-parameter Firth estimates of first reported installed-capacity entry. Points are odds ratios; bars are 95% intervals from 1,999 lineage bootstraps. Capacity compares 300 with 100 t/day; age is per ten years. The frames contain 13,072–15,154 risk rows; the legend gives events. Associations are not causal.

The supported conclusion is narrow but robust. First reported capacity entry is rare and selectively concentrated among larger facilities, and the association survives every continuity frame and event-level attack. It remains observational, not an effect of changing t/day.

4.3 RQ3: cohort differences are expressed mainly through reported sizing

The engineering-valid generator frame has 6,511 observations from 493 stable lineages. Median gross intensity is 0.327 MWh/t, median generator design intensity is 14.0 kW per t/day, median electrical capacity factor is 0.607, and median waste-processing utilization is 0.609. These summaries describe generator-years that pass stated bounds, not the full fleet.

Reported start-year cohorts differ much more in installed generator size than in annual capacity factor. The raw installed-kW model shows that, relative to 2010 or later and conditional on observed controls, adjusted installed capacity is 79.1% lower before 1990, 58.6% lower in 1990–1999, and 23.5% lower in 2000–2009.

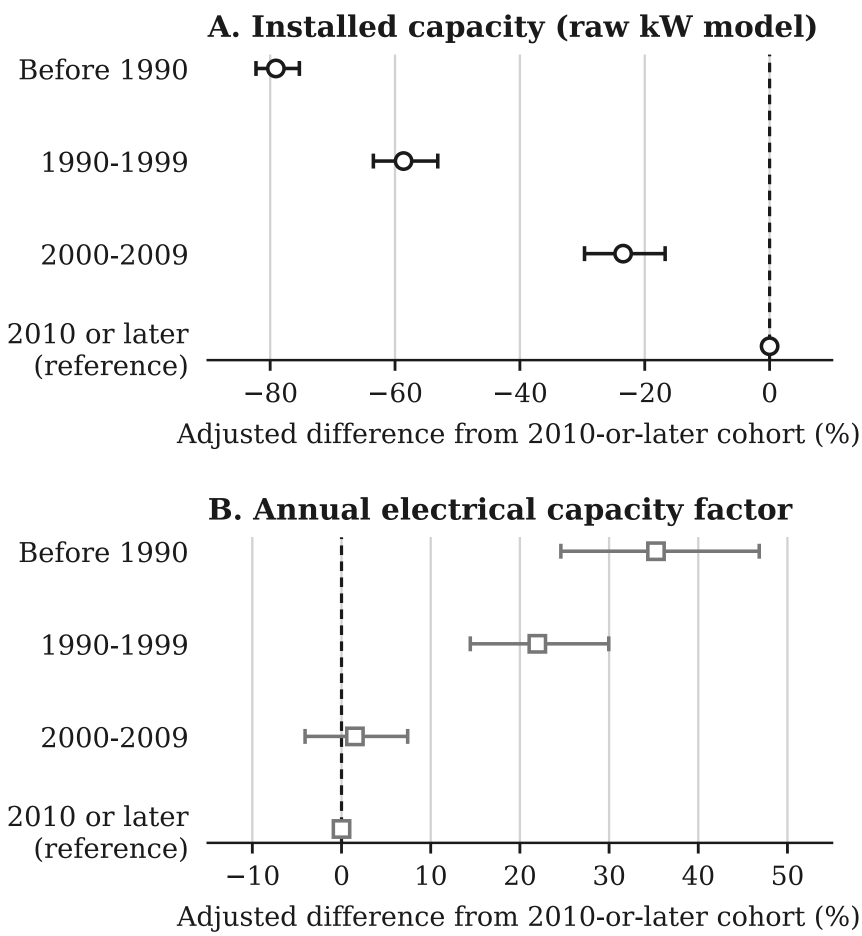

Figure 3: Adjusted start-year cohort contrasts for 6,511 facility-years from 493 lineages. Points are percentage differences from the 2010-or-later cohort; bars are lineage-clustered 95% confidence intervals. Models use the stated controls. Start year is not a verified installation date; contrasts are not causal.

The installed-kW elasticity with respect to processing t/day is 1.532 (95% CI 1.447–1.617), and model R^2 is 0.786. Under identical controls, subtracting one gives the 0.532 design-intensity elasticity; that is an algebraic change of outcome scale, not independent replication.

The electrical-capacity-factor model tells a different story. Conditional on processing scale, waste utilization, technology, furnace count, and fiscal year, pre-1990 and 1990s capacity factors are 35.3% (95% CI 24.6–46.8%) and 22.0% (14.4–30.0%) higher than in the 2010-or-later cohort. The 2000s estimate is 1.5% higher, with a CI from 4.1% lower to 7.4% higher. Waste utilization is positively associated with log electrical capacity factor (coefficient 1.695, 95% CI 1.448 to 1.942), while processing scale is negatively associated (-0.116, 95% CI -0.162 to -0.070). This model explains 33.9% of log capacity-factor variation. These coefficients describe annual use of installed kW. They do not show that utilization has an independent positive association with gross MWh/t after generator sizing is represented.

Under one common control design, the pre-1990 log gross-intensity gap of -1.250 decomposes into -1.565 from generator design, +0.016 from capacity factor, and +0.299 from lower waste utilization. The corresponding design, factor, negative-utilization, and total terms are -0.883, +0.020, +0.172, and -0.690 for the 1990s, and -0.267, -0.094, +0.089, and -0.272 for the 2000s. Sizing is the largest component in every older cohort. Excluding 14 cohort-switching lineages leaves the ordering unchanged. These exact conditional sums are accounting attribution, not causal mediation.

The direct gross-output model is consistent with the component structure. The elasticity of annual gross MWh with respect to throughput is 0.638 (95% CI 0.536 to 0.740), and the elasticity with respect to installed electrical capacity is 0.576 (95% CI 0.502 to 0.650), conditional on cohort, observed technology, furnace count, and year. The model R^2 is 0.914. It confirms that both waste loading and installed kW are central to annual output, but it is not a production function with exogenous inputs.

Because the components are algebraically coupled, a specification diagnostic compares gross-intensity models on the 5,806-row plausible-heating-value subset. In the legacy-style model without generator design intensity, the reported-age coefficient is -0.0349, the processing-capacity coefficient is 0.1001, and the waste-utilization coefficient is 0.6699; all have $p < 0.001$. After adding log generator design intensity, the corresponding coefficients become -0.0020 ($p = 0.2977$), -0.0092 ($p = 0.1991$), and -0.0995 ($p = 0.2038$). The sizing coefficient is 0.7532 ($p < 0.001$). This is not independent evidence of a mechanism or a causal decomposition: design intensity participates directly in the accounting identity. The diagnostic shows that the former age, scale, and utilization interpretation is specification-dependent once installed sizing is represented.

Adjacent-year rank persistence is highest for generator design intensity: 0.995 across 5,963 pairs from 470 lineages. Gross-intensity rank persistence is 0.961 and capacity-factor persistence is 0.873. The corresponding within-to-total variance ratios are 0.016 for log design intensity, 0.089 for log gross intensity, and 0.426 for log capacity factor. Sizing is therefore mostly a between-asset design attribute, while annual capacity factor contains much more within-asset movement.

The principal component conclusions remain stable across the two ten-year windows, lineage-equal weighting, and conservative or broad engineering bounds. For example, the processing-scale coefficient in the design-intensity model is 0.474 in FY2005–FY2014 and 0.577 in FY2015–FY2024; it is 0.520 under conservative bounds and 0.536 under broad bounds. Excluding every identity-uncertain lineage leaves 6,450 rows across 487 lineages and gives a nearly unchanged coefficient of 0.533. Within-asset-episode models also retain a positive association between utilization and electrical capacity factor. Those checks strengthen the component description but do not resolve simultaneous changes in throughput, maintenance, installed capacity, and output.

5 Discussion

5.1 What the paper changes in the fleet narrative

The 41.1% analytical participation rate describes equipment distribution, not waste coverage. Positive-output facilities handle 80.1% of throughput, and the installed segment holds 70.5% of processing design capacity. The all-record endpoint increase is 19.50 points but only 2.19 among endpoint-common lineages, so prevalence, composition, and volume coverage are distinct fleet facts.

Entry is strongly associated with processing scale in all four reduced models, and survives every event reclassification and whole-event-lineage deletion. The model still does not show that enlarging a facility would produce entry. Scale can proxy for unobserved municipal, technical, and financial conditions. Age is weaker: broad, prior-operation, and identity-certain bootstrap intervals span zero, while the same-episode result uses only 24 events. This sensitivity is evidence against a simple age-only narrative, not evidence for one preferred continuity definition.

Gross MWh/t also changes meaning after decomposition. Shared-control component regressions show that installed sizing is the largest part of every older- cohort log gap, while the utilization-adjusted capacity-factor model answers a different conditional question. The contribution is therefore not "newer plants work better"; it is a measured separation of design, annual use, and waste loading before an operating interpretation is assigned.

5.2 Evidence-bound implications

Monitoring should report facility, throughput, and design-capacity coverage together. Screening can treat processing scale as a marker for further feasibility work, but not use an age-only rule. Existing generators should be compared conditional on sizing and design vintage before gross MWh/t is interpreted as an operating gap.

These are measurement and diagnostic implications, not a ranked list of projects. The study does not determine whether a municipality should build, replace, coordinate, maintain, or retire a facility. Such decisions require capital costs, waste-supply agreements, grid and internal-use conditions, maintenance history, emissions controls, heat demand, and alternatives higher in the waste hierarchy. The paper's role is to prevent an aggregate statistic or misspecified ratio from deciding those questions implicitly.

5.3 Limitations and next evidence needed

The 20 workbooks are hash-identified and parser mappings are documented, but their original retrieval timestamps and publisher-side revision history are unavailable. Future acquisition should archive both.

Stable administrative lineages remain inferred. The resolver addresses exact duplicates, code breaks, row-order dependence, and known difficult links, but no physical-site registry confirms ownership, construction, or closure histories. Administrative absence is therefore not modeled as a physical outcome.

Entry is rare and partially left-censored. Firth estimation and lineage bootstrapping cannot overcome only 35 broad-frame, 33 prior-operation, and 24 same-episode events. Although all 1,999 requested bootstrap replications per frame converge, resampling cannot create missing project information. The parsimonious models omit financing, prices, contracts, catchments, maintenance, and detailed technology. Scale may proxy for these, so odds ratios are associations rather than effects of changing t/day.

Generator fields also have strict boundaries. Gross generation is not net export; on-site use and useful heat are incomplete; reported capacity may differ from availability; and MWh/t is not the European Union R1 energy-efficiency indicator, a lifecycle measure, or an economic measure (Grosso et al., 2010). Heating-value coverage is insufficient for a common thermal measure. Bounds remove 149 rows but may exclude unusual valid cases or retain plausible-looking errors.

Reported start year is not an equipment date and can bundle original design, later equipment, reporting, waste, and municipality context. Annual throughput, capacity factor, maintenance, and output are jointly determined. Controls, fixed effects, and adjacent differences do not create exogenous variation; the equations remain accounting identities and conditional descriptions.

The 2010-or-later cohort enters only from FY2010 onward. Fiscal-year indicators create within-year comparisons where cohorts overlap, but cannot remove selective survival, unrecorded replacement, or cohort-specific reporting.

Linking procurement, construction, generator, net-export, heat-use, outage, and waste-composition records would enable the project-specific questions this panel cannot answer. Exploratory administrative pathway comparisons remain in the supplement because their small selected groups do not support a main-text mechanism claim.

6 Conclusion

Japan's municipal-incineration fleet looks different at three margins. In FY2024, installed capacity appears in 41.1% of facility records, while positive-output facilities handle 80.1% of throughput and installed-capacity facilities represent 70.5% of processing design capacity. At the coverage margin, the 19.50-point all-record rise falls to 2.19 points among endpoint- common lineages. At the transition margin, first reported entry is sparse but strongly scale-selective: the broad 300-versus-100 t/day odds ratio is 6.72 and remains near seven under every frame and event-level attack. Age remains continuity-sensitive. At the component margin, common-control component sums identify installed sizing as the largest part of each older-cohort gross- intensity gap.

These findings reject two shortcuts without claiming a causal alternative. Facility count is not activity coverage, and newer-cohort gross MWh/t is not evidence of uniformly better operation. The transferable contribution is a three-margin diagnostic: reconstruct lineages before transitions, separate counts from covered activity, and separate installed design from annual use. That structure identifies which claim the administrative data can support and which next question requires project, engineering, cost, or causal evidence.

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Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey, which is publicly released by the Ministry and e-Stat. Analysis code, machine-readable result manifests, derived tables, figure-generation scripts, and manuscript figures are available in the versioned public repository at `https://github.com/Pann13223029/incineration-paper`. The repository preserves the source workbooks, cites e-Stat, and labels harmonized data as researcher-edited content. e-Stat permits reuse and modification with source citation under terms compatible with Creative Commons Attribution 4.0.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this manuscript, the author used OpenAI Codex and Anthropic Claude for language revision, manuscript organization, and assistance with code development and review. The author executed the analyses, inspected the source data and generated outputs, independently checked the reported results, reviewed and edited all AI-assisted material, and takes full responsibility for the content. These tools were not used as authors and did not replace author judgment or accountability.

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